Land-Use Classification

from Satellite Imagery Using CNNs

A comparison of custom CNNs and transfer learning, with evaluation of robustness to real-world image degradations.

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The Problem

Can deep learning classify terrain from overhead imagery?

Land-use classification from satellite imagery assigns each image a land-cover label — forest, residential, industrial, water, and so on.

Real-world applications:

- Environmental monitoring and land-use change detection
- Urban planning and infrastructure analysis
- Agricultural yield and crop-type estimation
- **Autonomous aerial navigation in GPS-denied environments**

Why aerial navigation?

A drone operating without GPS must recognize the terrain below to navigate. Real-time classification of what it sees — forest, road, water — is the foundation of visual odometry.

GPS denial is no longer hypothetical: intentional jamming and spoofing are routine in contested airspace, and natural occlusions like canyons or dense urban areas routinely degrade GPS even in civilian use.

This motivates evaluating not just clean-data accuracy, but robustness under noise, blur, low light, and arbitrary headings.

Dataset: EuroSAT

27,000 Sentinel-2 satellite images across 10 land-cover classes

27,000	10	64×64	10 m / px
Labeled images	Balanced classes	Native pixels (Sentinel-2 RGB)	Ground resolution

Train / Validation / Test split: 70% / 15% / 15% (stratified, fixed seed 42). Training augmentation: horizontal flip + ±15° rotation.

AnnualCrop

Forest

HerbaceousVegetation

Highway

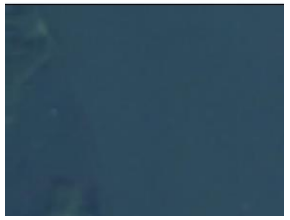
Industrial

Pasture

PermanentCrop

Residential

River


SeaLake

Research Questions

What this project sets out to answer

1

Baseline capability

Can a small custom CNN, trained from scratch, achieve reasonable accuracy on satellite imagery?

2

Transfer learning value

How much does transfer learning from ImageNet help — frozen vs. fine-tuned?

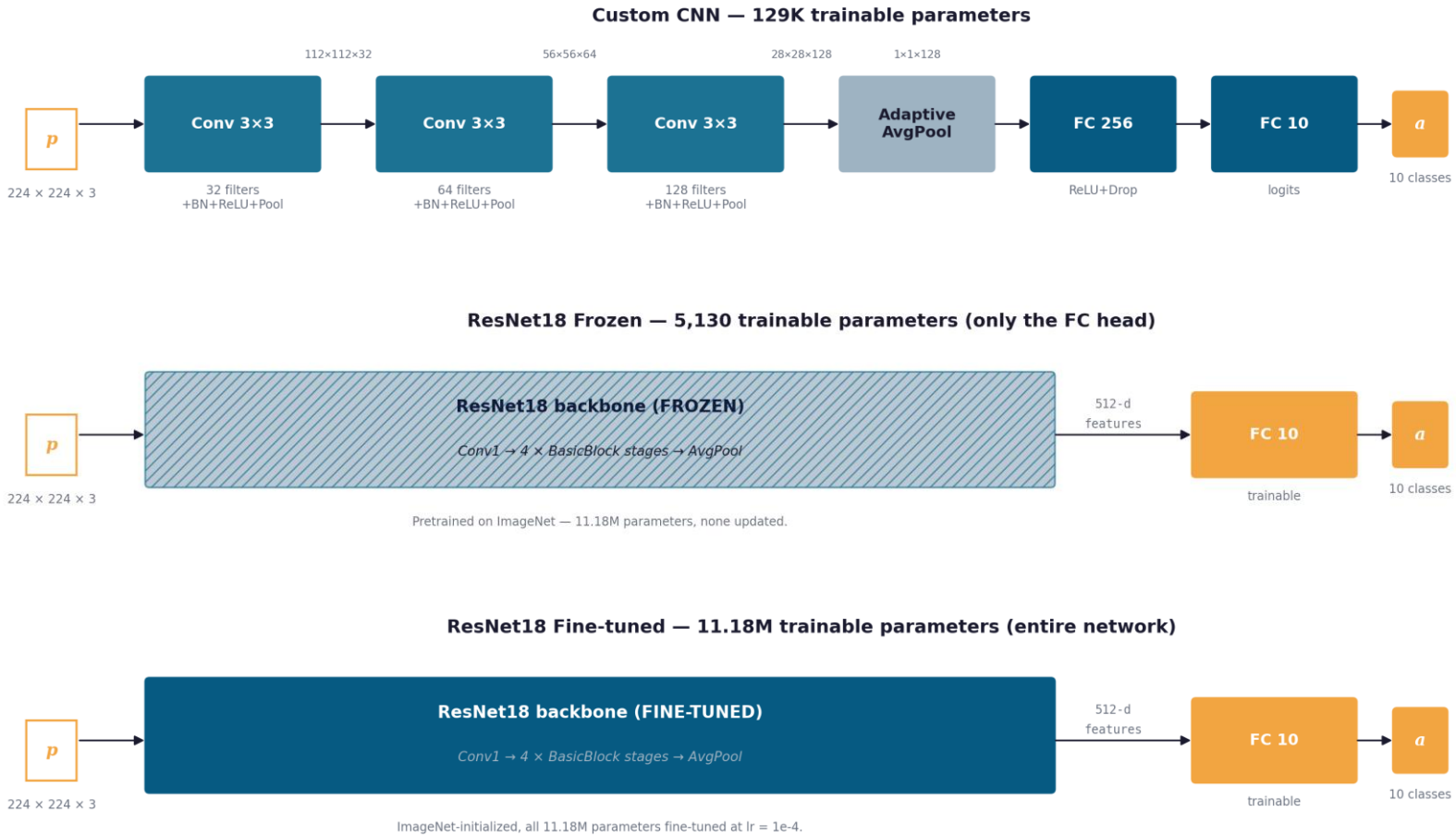
3

Real-world robustness

How do these models degrade under noise, blur, resolution loss, rotation, and lighting shifts?

Three Models, Same Pipeline

Increasing reliance on pretrained knowledge



Shared training pipeline: PyTorch · CrossEntropyLoss · Adam (lr 1e-3 baseline & frozen, lr 1e-4 fine-tuned) · 25 epochs · batch size 64 · identical 70/15/15 splits

Methodological Pivot: Resolution

Why training at 224×224 was the right call

Initial training at 64×64 (native)

Frozen ResNet only reached **82%** — well below baseline expectations for transfer learning.

Hypothesis: ResNet18 was designed for 224×224 input.

At 64×64, its deepest feature map collapses to 2×2 spatial resolution — far smaller than what the architecture was designed around.

Resolution: all inputs upsampled to 224×224. The baseline CNN uses adaptive average pooling, allowing one architecture to accept both input sizes without rewriting the FC layer dimensions.

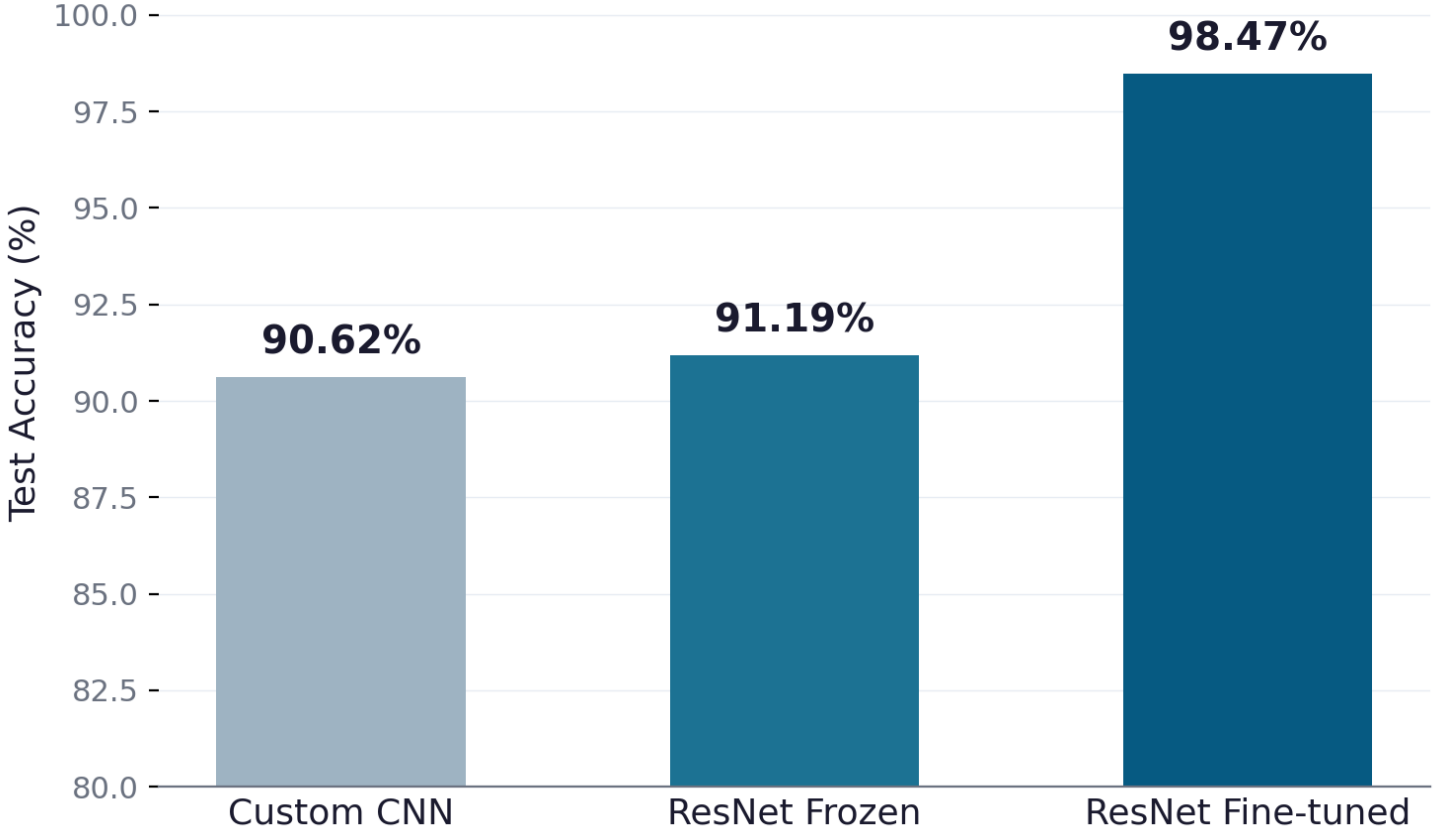
Before vs. After retraining

Model	64×64	224×224
Baseline CNN	92.59%	90.62%
ResNet frozen	81.98%	**91.19%**
ResNet fine-tuned	97.41%	98.47%

Frozen ResNet's +9% confirmed the hypothesis — pretrained features perform substantially better at native resolution.

Clean Test Results

Fine-tuning delivers a decisive accuracy jump



Key observations

Baseline CNN essentially ties frozen ResNet

A 129K-parameter network matches a 5K-parameter classifier on ImageNet features. Either approach gets you ~91%.

Fine-tuning adds +7 percentage points

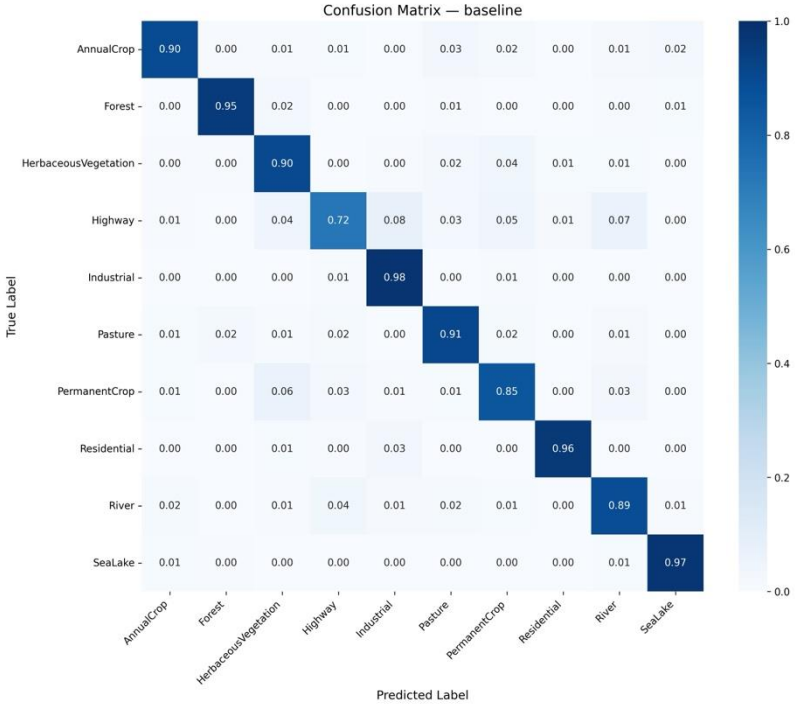
Adapting ImageNet features to the satellite domain closes nearly all remaining error. 98.47% is near the dataset ceiling.

So — is fine-tuning always the right call?

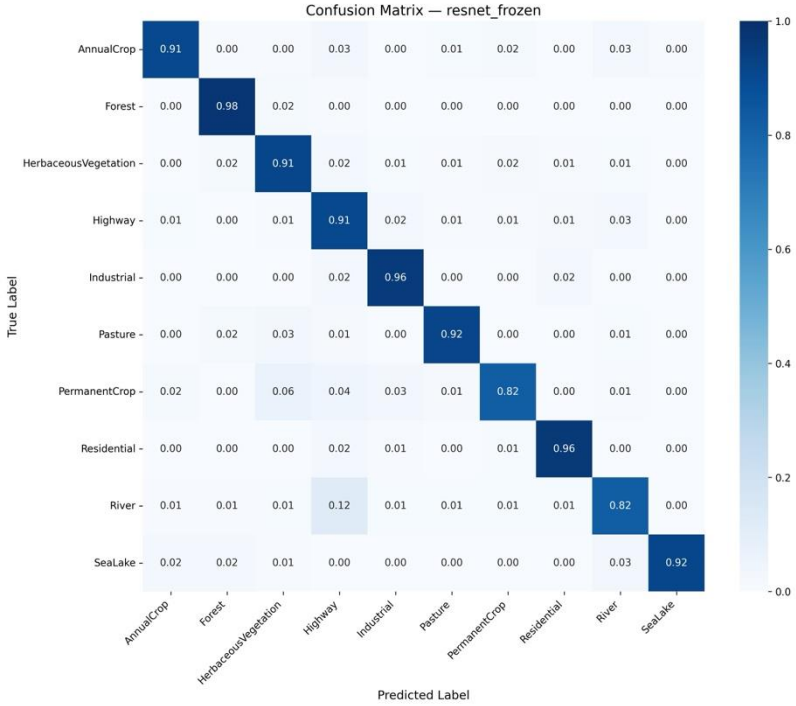
What Errors Remain?

Confusion matrices reveal where each model struggles

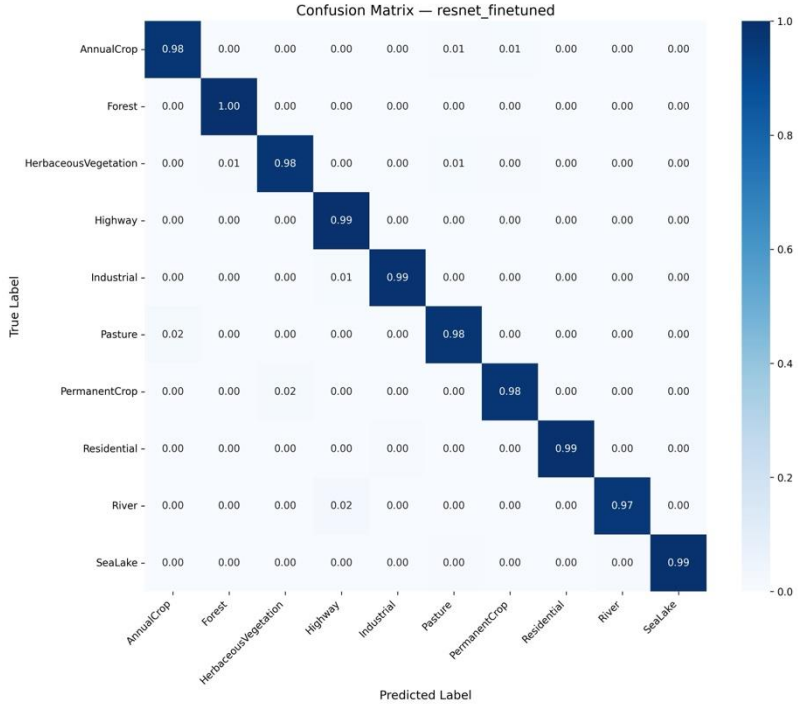
Custom CNN (90.62%)



ResNet Frozen (91.19%)



ResNet Fine-tuned (98.47%)

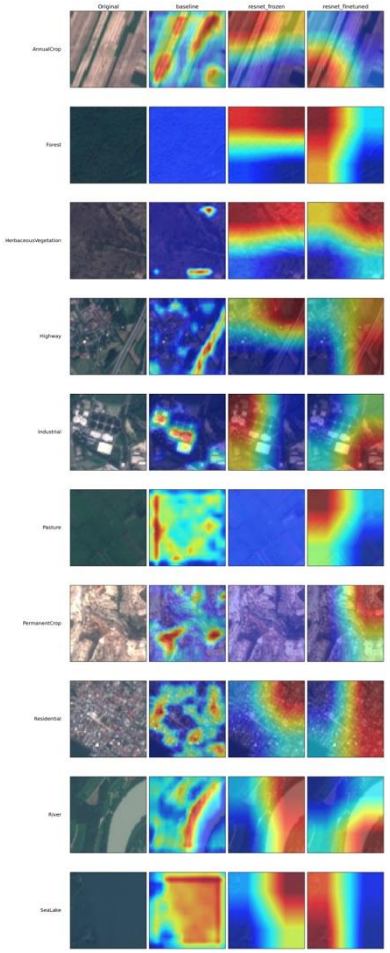


Baseline: struggles with Highway (72%) — elongated linear features confused with rivers and urban edges.
Frozen: 12% of Rivers misclassified as Highways — ImageNet features can't distinguish water from asphalt.
Fine-tuned: River → Highway drops from 12% to 2%. Fine-tuning learned the satellite-specific features that resolve this ambiguity.

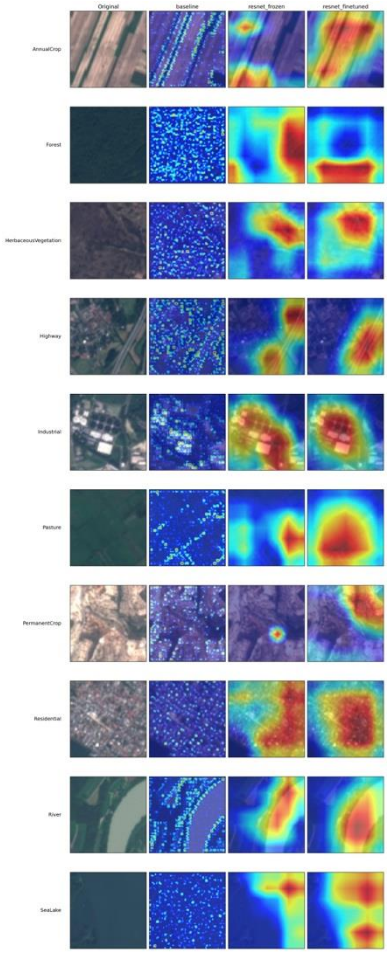
What Is the Model Looking At?

Grad-CAM heatmaps: 64×64 input vs. 224×224 input

64×64 input (initial training)



224×224 input (final)



The interpretability gap

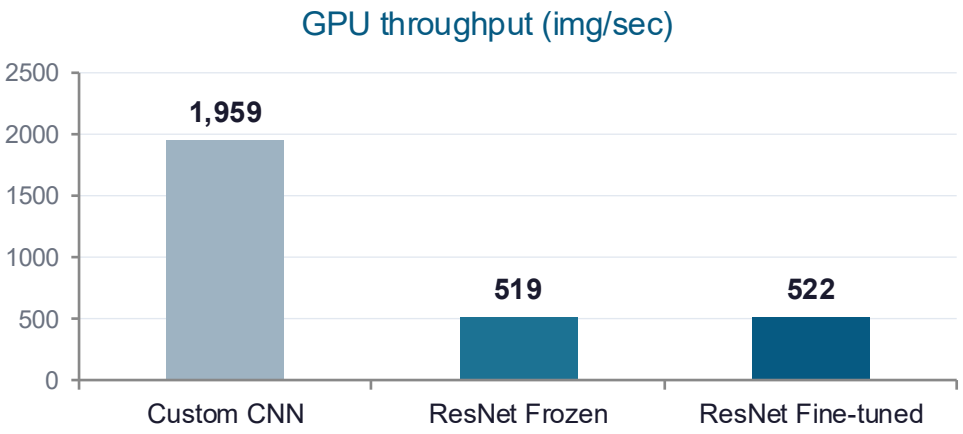
Resolution matters: at 64×64 the ResNet's last conv layer produces only 2×2 feature maps, so heatmaps are blurred to a few large blobs. At 224×224 it has 7×7 maps, and attention becomes more spatially specific.

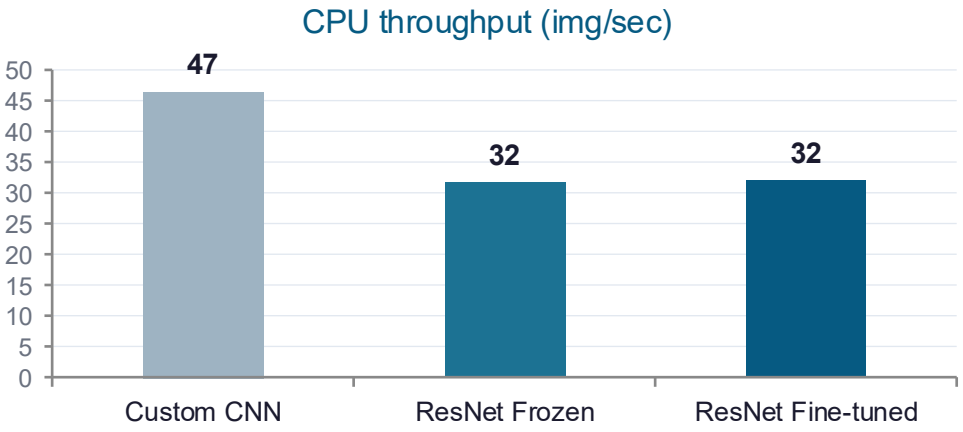
Why the baseline looks different from ResNet: the baseline CNN has 8×8 maps even at 64×64 input (only 3 pooling stages vs. ResNet's 5). It also has only 128 channels of relatively low-level features, so its heatmaps highlight edges and small textural cues. ResNet has 512 deep, semantic channels — its heatmaps reflect distributed, high-level patterns.

Sharper heatmaps ≠ better generalization. The baseline's clean blobs come from low-level features. The fine-tuned ResNet's diffuse heatmaps reflect the kind of distributed, high-level reasoning that achieves 98.5% accuracy.

How Fast Are They?

Inference latency and throughput on CPU and GPU





Model	Device	Latency (ms)	Throughput	Params
Custom CNN	GPU	**0.51 ± 0.02**	**1,959 img/s**	129K
Custom CNN	CPU	21.5 ± 3.1	**46.6 img/s**	129K
ResNet Frozen	GPU	1.93 ± 0.05	519 img/s	11.2M
ResNet Frozen	CPU	31.3 ± 1.4	31.9 img/s	11.2M
ResNet Fine-tuned	GPU	1.92 ± 0.04	522 img/s	11.2M
ResNet Fine-tuned	CPU	31.1 ± 0.8	32.1 img/s	11.2M

Observations

- Custom CNN on GPU runs at nearly 2,000 img/s — 3.8× the throughput of either ResNet.
- Frozen and fine-tuned ResNets are identical at inference (same architecture, same parameter count).
- On CPU the gap closes (~1.5×) because both models become memory-bandwidth bound.
- **For drone deployment:** the baseline CNN delivers 90.6% accuracy at 4× lower latency — a tradeoff worth considering on edge hardware.

Robustness Evaluation

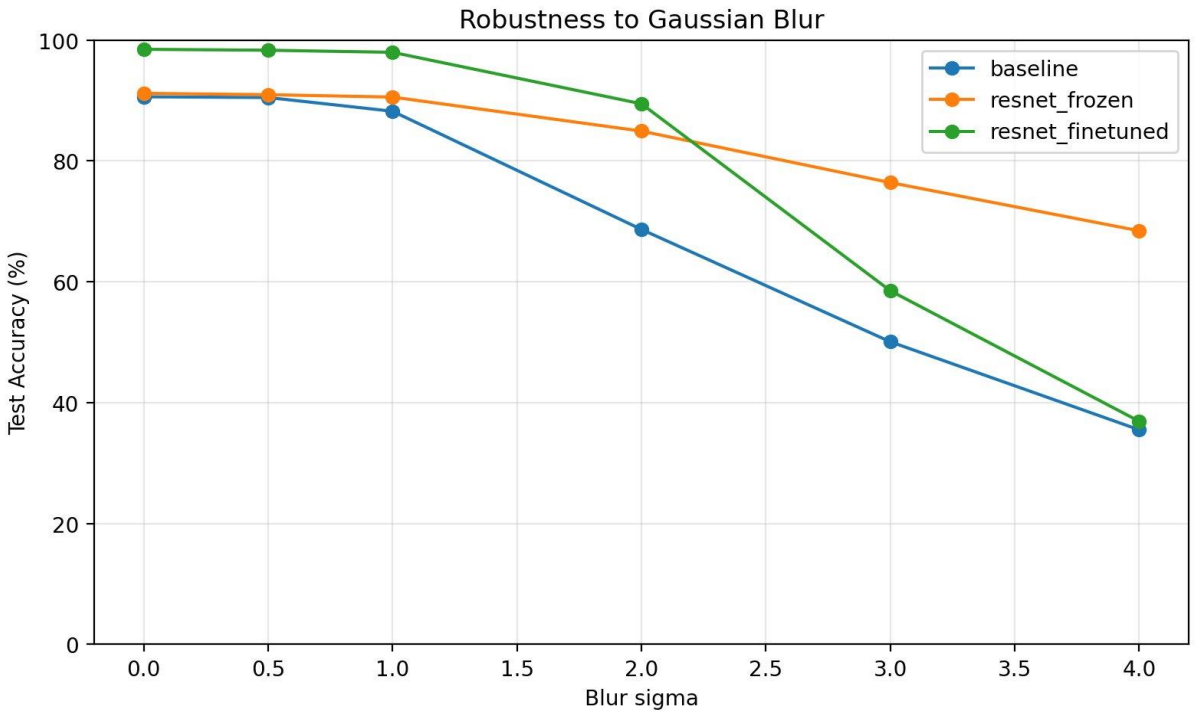
Five degradations simulating real-world aerial imagery

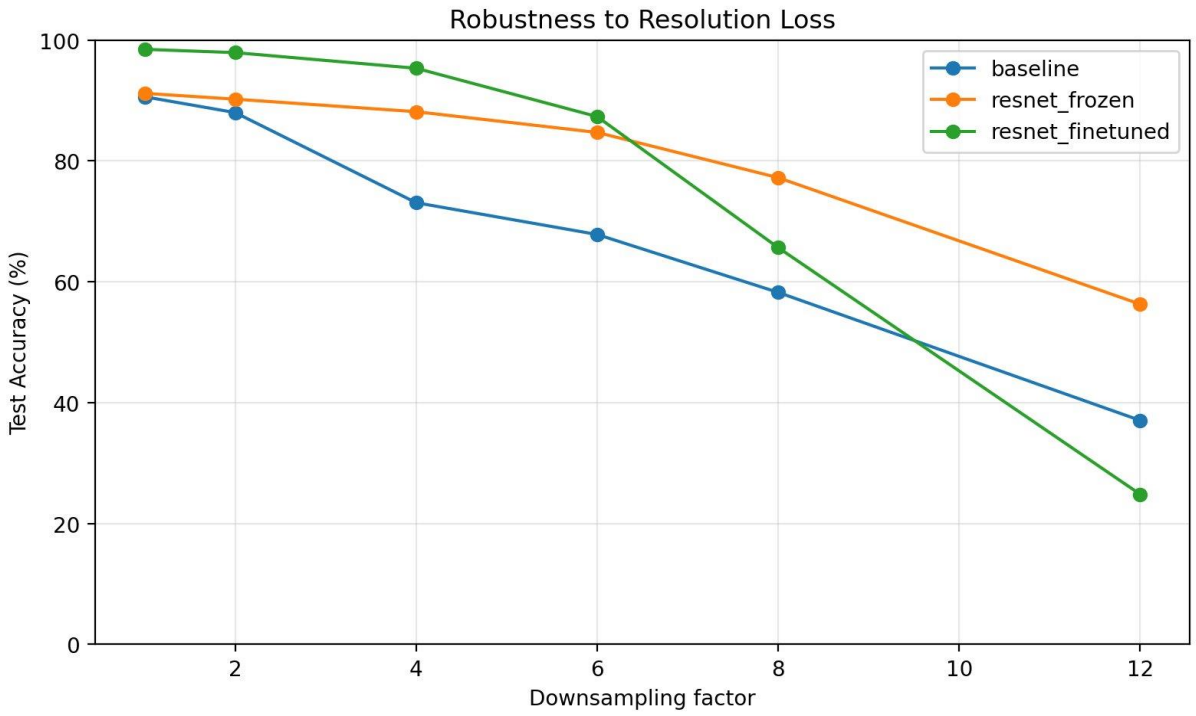
Clean accuracy is not enough. A deployed aerial system encounters sensor noise, motion blur, lower-resolution optics, arbitrary flight headings, and varying lighting throughout the day.

1	**Gaussian noise**	Sensor noise in low-light or low-cost cameras	$\sigma = 0.00 \rightarrow 0.30$
2	**Gaussian blur**	Motion blur, atmospheric haze, or defocus	$\sigma = 0.0 \rightarrow 4.0$
3	**Resolution loss**	Lower-quality sensors or extreme altitude	$factor = 1\times \rightarrow 12\times$
4	**Rotation**	Drone approaching terrain from arbitrary heading	$angle = 0^\circ \rightarrow 270^\circ$
5	**Brightness**	Time-of-day and exposure variation	$multiplier = 0.25 \rightarrow 1.5$

The Fragility Finding

Fine-tuning trades robustness for accuracy



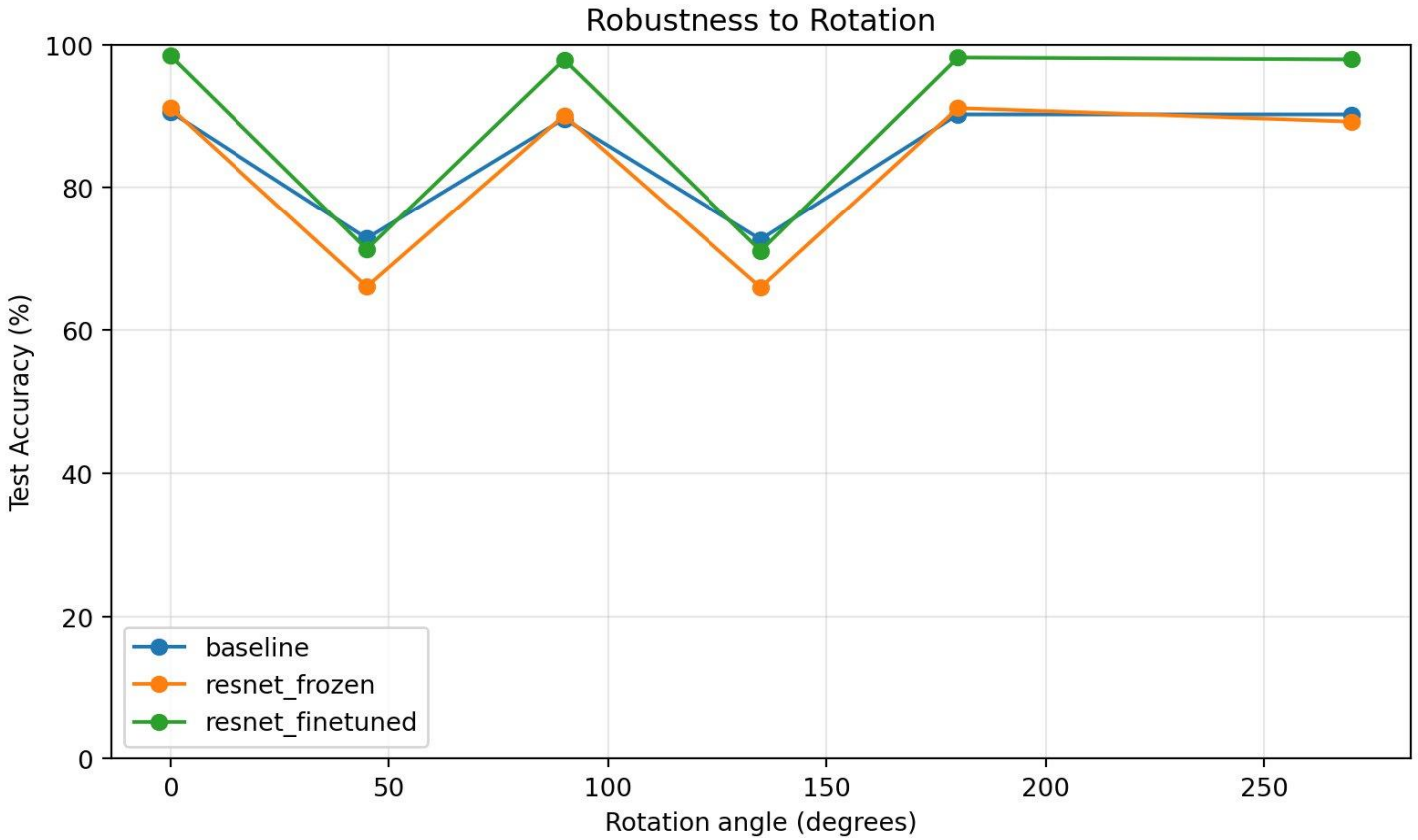


Crossover effect:

Fine-tuned ResNet dominates on clean and mildly degraded data, but frozen ResNet is more robust at severe blur (σ=4) and extreme downsampling (factor 12). Specialized features generalize worse than generic ImageNet features when conditions fall far from the training distribution.

The Rotation Zigzag

Models learned 90° rotations, not true rotation invariance



A clean artifact of training

Accuracy peaks at 0°, 90°, 180°, 270° — and drops ~25 points at 45° and 135°.

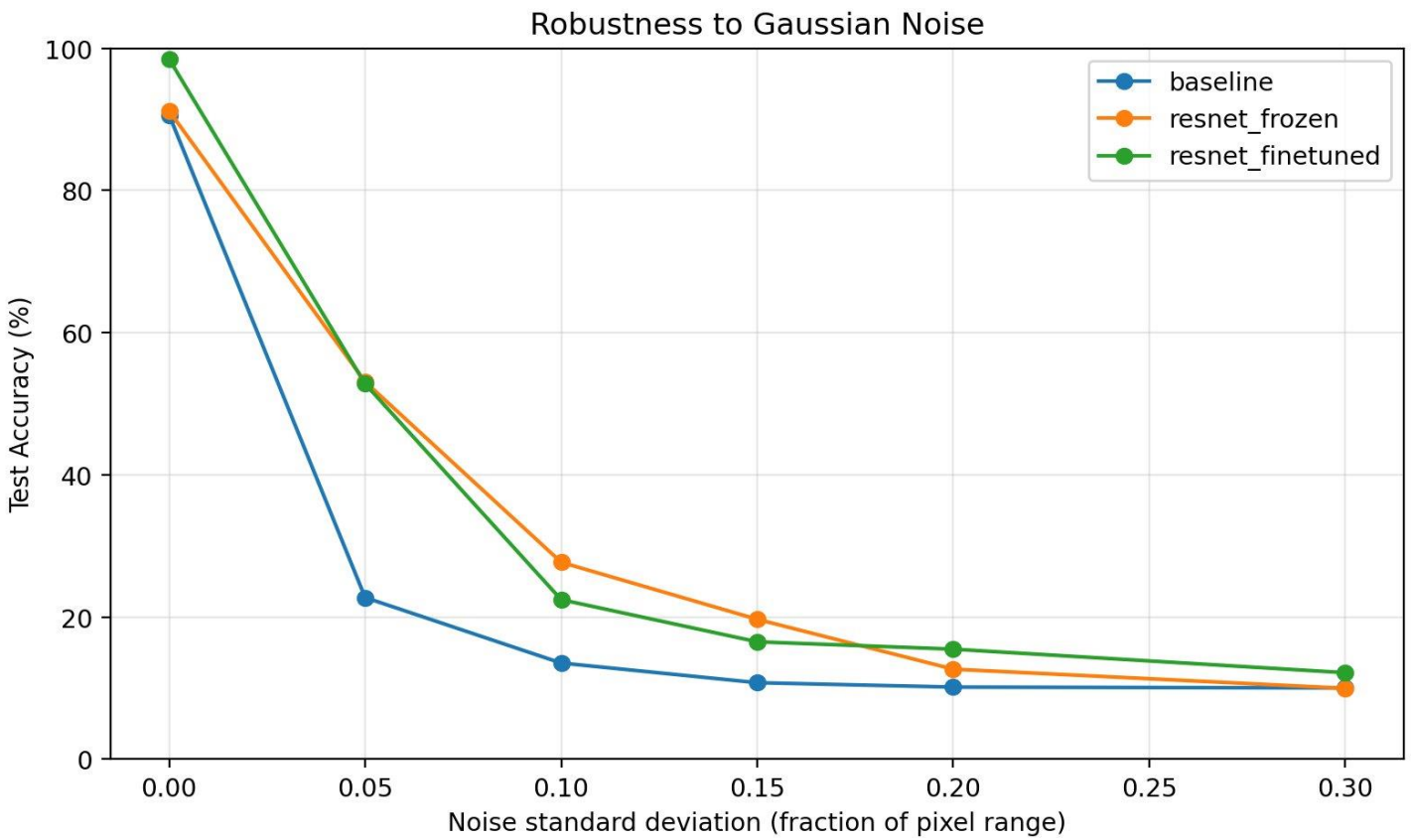
Why? Training augmentation included ±15° rotation and random horizontal flips, so the model effectively saw cardinal-aligned imagery only.

The model did not learn rotation invariance — it memorized that satellite features appear aligned to the grid axes.

Practical implication: this is a training-data limitation. A rotation-robust classifier would require expanding augmentation to cover the full 0–360° range, or training on a more diverse dataset that includes off-axis imagery.

Fragility Under Noise

All three models collapse immediately



No defense without training

At σ = 0.05 (mild noise), all three models drop from >90% to ~52%.

By σ = 0.15, accuracy is at chance (10% for 10 classes).

Takeaway: robustness does not come for free. It must be explicitly trained for. Clean-data accuracy is an incomplete measure of real-world performance.

Fix: add noise and lighting augmentation to the training pipeline (future work).

Conclusions

What we learned — and what it means for deployment

Fine-tuning wins on clean data

98.5% test accuracy. +7 points over frozen transfer learning. Satellite imagery benefits substantially from domain adaptation.

The custom CNN is surprisingly competitive

At 90.6%, a simple 129K-parameter network matches frozen ResNet — and runs ~3.8× faster on GPU and ~1.5× faster on CPU.

Clean accuracy does not predict robustness

The most accurate model can also be the most fragile. Fine-tuned features are brittle under severe blur and resolution loss.

Augmentation defines what models can handle

The rotation zigzag and noise collapse are direct artifacts of what was (or wasn't) in the training augmentation pipeline.

Future Work & References

Where this goes next

Future work

- Add noise, blur, and lighting augmentation to training; re-evaluate the robustness curves
- Test cross-geographic generalization — EuroSAT (Europe) trained, evaluated on U.S. imagery
- Benchmark inference speed on edge hardware (Jetson Nano, Raspberry Pi)
- Evaluate on real drone imagery rather than degraded satellite data
- Explore architectures explicitly designed for rotation invariance (e.g., equivariant CNNs)

References

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Thank you — questions?